

Question Bank



**MALLA REDDY
UNIVERSITY**

(Telangana State Private Universities Act No.13 of 2020 and
G.O.Ms.No.14, Higher Education (UE) Department)

Deep Learning

III B.Tech I Semester

(2025-26)

Department of CSE (Data Science)

School of Engineering



School of Engineering-A.Y 2025-2026 Regulation: R23
(CSE-DATA SCIENCE)
III Year B. Tech - I Semester
Subject Name: Deep Learning
Subject Code: MR23-1CS0332

QUESTION BANK

Qno	Question	Marks	Section
1	Explain the role of sigmoid neurons in feed forward neural networks.	8	Section-I
2	What is a perceptron, and how does the perceptron learning algorithm work?	8	Section-I
3	a) What are multilayer perceptrons (MLPs)? b) How do they differ from single-layer perceptrons?	4 4	Section-I
4	How do feed forward neural networks achieve their representation power?	8	Section-I
5	a) What is back propagation? b) How is it used to train feed forward neural networks?	4 4	Section-I
6	Write the differences between Feed Forward and Feed Backward Neural networks.	8	Section-I
7	What is Deep learning? Explain its uses and applications.	8	Section-I
8	Discuss about single layer perceptions in detail.	8	Section-I
9	Differentiate between machine learning and deep learning in detail with examples.	8	Section-I
10	How does the McCulloch-Pitts neuron model capture the basic functionality of a biological neuron, and what are the key components and processes involved in its operation?	8	Section-I
11	What is the difference between Stochastic Gradient Descent and Batch Gradient Descent?	8	Section-II
12	a. How does Momentum Based Gradient Descent work? b. How is it different from other optimization algorithms?	4 4	Section-II
13	How does basis transformation using principal component analysis (PCA) help in reducing the dimensionality of data in deep learning?	8	Section-II
14	Explain the Adam optimization algorithm in detail.	8	Section-II
15	How do the various optimization algorithms (GD, Momentum-based GD, Nesterov accelerated GD, AdaGrad, RMSProp, Adam) impact the convergence rate?	8	Section-II
16	Write about RMS optimization algorithm in detail.	8	Section-II
17	Explain the concept of singular value decomposition (SVD) and how it is used in deep learning.	8	Section-II
18	How are eigenvalues and eigenvectors utilized in deep learning, and what significance do they hold in the analysis of matrices, particularly in tasks such as principal component	8	Section-II

School of Engineering-A.Y 2025-2026 Regulation: R23
(CSE-DATA SCIENCE)
III Year B. Tech - I Semester
Subject Name: Deep Learning
Subject Code: MR23-1CS0332

QUESTION BANK

	analysis and Understanding the behavior of neural network layers?		
19	How does eigenvalue decomposition contribute to the analysis of gradient descent in deep learning?	8	Section-II
20	Explain in detail about the concept of gradient based learning.	8	Section-II
21	What are autoencoders and how do they relate to principal component analysis (PCA) in terms of dimensionality reduction?	8	Section-III
22	a) How do sparse autoencoders differ from traditional autoencoders? b) What are some of the applications of sparse autoencoders in deep learning?	4 4	Section-III
23	How does bias-variance trade-off can be managed using Regularization techniques such as L2 regularization?	8	Section-III
24	How do the practices of introducing noise at the input and implementing the dropout regularization technique contribute to mitigating overfitting and improving the generalization performance of a model? Explore the trade-offs, advantages, and limitations associated with each approach, and offer instances where these methods have proven to be effective in various applications.	8	Section-III
25	Explore the concept of dataset augmentation in deep learning and how it contributes to improving model generalization.	8	Section-III
26	What is the role of denoising autoencoders in deep learning, and how do they enhance the robustness of a model?	8	Section-III
27	Explain the concept of contractive autoencoders and how they differ from traditional autoencoders.	8	Section-III
28	Describe the concepts of parameter sharing and tying in the context of autoencoders.	8	Section-III
29	Discuss the concept of early stopping as a regularization technique in deep learning.	8	Section-III
30	Compare and contrast the effects of injecting noise at the input layer and using dropout regularization in deep learning models.	8	Section-III
31	a) What are some of the key advantages and limitations of popular activation functions? b) How can we develop better activation functions that are more effective for specific applications?	4 4	Section-IV
32	a) What are Learning Vectorial Representations of Words?	4	Section-IV

School of Engineering-A.Y 2025-2026 Regulation: R23
(CSE-DATA SCIENCE)
III Year B. Tech - I Semester
Subject Name: Deep Learning
Subject Code: MR23-1CS0332

QUESTION BANK

	b) How can techniques such as Word2Vec be used to generate high-quality word embedding that capture the semantic relationships between different words in a corpus?	4	
33	a) Solve the concept of greedy layer wise pre-training? b) How it is used to improve the training of deep neural networks?	4 4	Section-IV
34	How can we use advanced weight initialization methods to improve the performance of Deep Learning models?	8	Section-IV
35	How does Batch Normalization work, and how can we use this technique to improve the training stability and generalization performance of Deep Learning models?	8	Section-IV
36	Discuss scenarios where Greedy Layer wise Pre-training is particularly advantageous and its impact on model convergence.	8	Section-IV
37	Discuss the significance of using advanced activation functions in neural networks. How do alternatives like ReLU and variants address the limitations of traditional functions?	8	Section-IV
38	a) Explain backpropagation Through Time (BPTT). b) Write briefly about the limitations of BPTT.	4 4	Section-IV
39	Describe the process by which the Batch Normalization technique normalizes activations within intermediate layers during training.	8	Section-IV
40	How does the Word2Vec algorithm contribute to learning vector representations of words? Explain the underlying principles of continuous bag-of-words (CBOW) and skip-gram models within Word2Vec. Discuss the practical applications and benefits of utilizing Word2Vec.	8	Section-IV
41	How can Convolutional Neural Networks (CNNs) be used to extract high-level features from image and video data, and what are some of the key technical challenges involved in this process?	8	Section-V
42	What are Gated Recurrent Units (GRUs) and Long Short-Term Memory (LSTM) cells, and how can these specialized RNN architectures be used to model long-term dependencies and avoid the vanishing gradient problem?	8	Section-V
43	Explore the GoogLeNet architecture and elucidate some of its significant contributions to the field of deep learning.	8	Section-V
44	a) Explain Long Short term Memory (LSTM)?	4	Section-V

School of Engineering-A.Y 2025-2026 Regulation: R23
(CSE-DATA SCIENCE)
III Year B. Tech - I Semester
Subject Name: Deep Learning
Subject Code: MR23-1CS0332

QUESTION BANK

	b) What are some of the key technical challenges involved in this process?	4	
45	Examine the architecture of AlexNet in the realm of deep learning. Highlight key aspects that have significantly impacted the field, showcasing the model's advancements and innovations.	8	Section-V
46	How can Recurrent Neural Networks (RNNs) and Back propagation through time (BPTT) be used to model temporal dependencies in sequential data such as speech, text and music?	8	Section-V
47	Examine the ResNet architecture and highlight key contributions it has made to the field of deep learning.	8	Section-V
48	a) Explore the applications of Deep Dream and Deep Art in the field of deep learning.	4	Section-V
	b) How do these techniques leverage neural networks to generate visually captivating and artistic images?	4	
49	How does Guided Backpropagation enhance interpretability in deep neural networks? Discuss the key principles behind this technique.	8	Section-V
50	a) Explain the concept of attention mechanisms in neural networks.	4	Section-V
	b) How do attention mechanisms are used over images.	4	